

“I’m” Lost in Translation: Pronoun Missteps in Crowdsourced Data Sets

Katie Seaborn
Tokyo Institute of Technology
seaborn.k.aa@m.titech.ac.jp

Yeongdae Kim
Tokyo Institute of Technology
kim.y.ah@m.titech.ac.jp

ABSTRACT

As virtual assistants continue to be taken up globally, there is an ever-greater need for these speech-based systems to communicate naturally in a variety of languages. Crowdsourcing initiatives have focused on multilingual translation of big, open data sets for use in natural language processing (NLP). Yet, language translation is often not one-to-one, and biases can trickle in. In this late-breaking work, we focus on the case of pronouns translated between English and Japanese in the crowdsourced Tatoeba database. We found that masculine pronoun biases were present overall, even though plurality in language was accounted for in other ways. Importantly, we detected biases in the translation process that reflect nuanced reactions to the presence of feminine, neutral, and/or non-binary pronouns. We raise the issue of translation bias for pronouns and offer a practical solution to embed plurality in NLP data sets.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Social and professional topics**; • **User Characteristics**; • **Gender**; • **Computing methodologies**; • **Artificial Intelligence**; • **Natural Language Processing**; • **Lexical semantics**;

KEYWORDS

Gender bias, Data sets, Natural language processing, Translation, Feminist HCI

ACM Reference Format:

Katie Seaborn and Yeongdae Kim. 2023. “I’m” Lost in Translation: Pronoun Missteps in Crowdsourced Data Sets. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3585667>

1 INTRODUCTION

Initiatives in natural language processing (NLP) for communication between people and virtual assistants (VAs) is taking off [9, 28, 36]. Many are gathering and offering large, crowdsourced data sets for this purpose [11]. An ongoing challenge is that most are in English and geared towards the US context [12, 20, 23]. In recognition of this, new efforts have started to be undertaken to ensure that a diversity of languages and cultural contexts are represented. For

example, the Amazon Alexa team have been spearheading a “massive” crowdsourced translation and localization initiative of their MASSIVE data set into 51 languages [19], including a global competition¹. Indeed, most of these initiatives rely on crowdsourcing activities. However, critical analyses have identified biases in translation, notably around gender representation in pronouns [32, 41] and word choice [10, 32, 38]. As such, alongside the move towards voice- and speech-based interaction [15, 34], gender bias in NLP is now a hot topic for feminist HCI work [3, 4, 6, 13, 32].

One issue that remains underexplored is how pronouns are *translated* language-to-language [14, 21, 30]. Languages vary greatly on pronoun use. For example, in English, a small set of pronouns are commonly used, but in Japanese, pronouns are less commonly used and arguably more varied [29]. We also need to consider gender ambiguous or “gendered ambiguous” pronouns that could represent more than one (or no) gender identity [41]. A common example is “they/them,” which can be interpreted as a neutral plural pronoun or a singular non-binary pronoun. Moreover, trans*, non-binary, and other genderqueer folk may use “gender neutral” pronouns like “they/them” or gendered pronouns like “he/him” and “she/him,” as well as a diversity of other pronouns, such as “ze/xe/xem/zem” [22, 25, 33, 35]. Translators need to make careful choices about how to translate pronouns in a case-by-case fashion, and also be aware of the cultural context of variable pronouns. However, crowdsourcing contributors are likely not trained in such matters, as evidenced by the small body of work on pronoun translation for NLP data sets so far [32] and when considering related areas, such as crowdsourcing efforts and user-generated content for machine translation [14, 21, 30, 42]. To the best of our knowledge, no work to date has looked at the context of NLP data sets geared towards VA training.

As a first step, we evaluated pronoun translations in *Tatoeba*², an open NLP database featuring crowdsourced translations between English and Japanese. We chose the case study of pronouns as an important identity, diversity, and inclusivity marker [25, 31, 33], as well as a pluralistic facet of language with implications for translation. Our overall research question (RQ) was: *Are there gender biases in pronouns within crowdsourced translations of English data sets to Japanese?* Our aim was to evaluate translation accuracy, so we first asked: *RQ1: Have pronouns been translated faithfully from English to Japanese?* We then focused on two important aspects when translating pronouns: (i) the relative number of pronouns linked to certain genders and (ii) whether one-to-one translations of pronouns were faithfully preserved. We were guided by long-standing language biases in English and romantic languages [5], recently confirmed within the latest crowdsourced English NLP data sets [8, 17, 18, 32] and may thus feature in translations of these data

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585667>

¹The Massively Multilingual NLU 2022 workshop (MMNLU-22 at EMNLP 2022): <https://mmnlu-22.github.io>

²<https://tatoeba.org>

sets to Japanese and other languages. As such, we asked specific questions about pronoun translation: *RQ2: Is there a masculine bias in pronouns across the data sets?* And *RQ3: Are there more instances of mistranslated masculine pronouns than other pronouns?* We aim to raise awareness of sociocultural biases that may be embedded in translation work and offer a technical solution to account for pronoun diversity and context-sensitive variability. We contribute our findings on the most recent and well-known NLP data sets used for training VA speech and a way to embed plurality in pronouns.

2 RELATED WORK

We situate our work at the intersection of translations of NLP data sets and social justice initiatives within HCI.

2.1 Gender Mistranslations in NLP

While virtually no work has considered gender translation biases in NLP data sets, a growing body of work has considered machine translation [14, 21, 30]. Google Translate, for instance, is a long-standing example of a gender biased system that relies on user-generated content; it specifically mistranslates pronouns and assigns masculine pronouns in the presence of ambiguous pronouns or a lack of pronouns [30, 42]. Moreover, recent examples of mass-scale translation efforts in NLP spaces, specifically for VAs, indicates that biases, including gender biases, have not been considered. For example, the authors of the MASSIVE data set do not mention bias in their report³. As such, we hypothesize:

H1. There will be mismatches in direct pronoun translations from English to Japanese.

2.2 Masculine Biases in NLP

A wealth of research on NLP and machine translation has pointed to gender biases and especially masculine biases [10, 12, 20, 32]. A recent word embeddings study of over 630 billion sources on the Internet evaluating the word “person” and “people” [2] found that these “gender neutral” words are often equated with “man” and “men.” Breaking work by Seaborn, Chandra, and Fabre [32] has revealed implicit masculine biases in MASSIVE as well as the older ReDial NLP data sets, designed for training VAs and conversational user interfaces. This is not unprecedented, nor localized to NLP; this is a human language and cultural phenomenon. For instance, when there are no gender markers, such as names or titles, or pronouns, people are given to assume the person is a man [1]. We thus hypothesize for Tatoeba:

H2. There will be a masculine bias in pronoun use in the data sets.

Moreover, given the literature on crowdsourcing and user-generated content for machine translation alongside likely gender biases, we would expect contributors who make translations to carry over biases from one language to another. Indeed, a growing number of critical voices within NLP spaces have raised awareness of gender and other biases that can creep into the creation process, from who created the data to who curated and annotated it [7, 27]. We expect this to be true for translation, which outside of NLP is recognized to be a process where biases come into play [14, 39],

including as a creative part of localizing and situating work within other cultures, i.e., transcreation [16]. As such, we hypothesize:

H3. There will be a masculine bias in pronoun translation from English to Japanese.

3 METHODS

We conducted an automated content analysis of pronoun use in English and Japanese within Tatoeba. For this, we used NLP methods to extract meaningful patterns about pronoun presence and use, especially in translation, from the text materials in the data sets. Our protocol was registered before data collection on January 15th, 2023 via OSF⁴.

3.1 Data Set

We used the Tatoeba database of crowdsourced sentences and their translations across 359 languages, specifically the English and Japanese data sets. The downloadable version of the database is updated every Saturday at 6:30 a.m. (UTC); we used the version from December 24th, 2022. While it is difficult to link translations to certain people, statistics indicate that contributors total 7,027 native⁵ and 13,558 users⁶ of English and 431 native and 2,997 users of Japanese. The data sets are made up of numbered rows, such that each row is paired with a row in the other language data set. Here is an example from the Japanese and English data sets:

Line 51: が最後に自分の考えをえた人は、をいだ
と思ったようだ。

Line 51: The last person I told my idea to thought I
was nuts.

Tatoeba was originally created by Trang Ho in 2006 and hosted on SourceForge as “multilangdict.” Coincidentally, “tatoeba” is a Japanese word that means “for example.” We chose Tatoeba for a few reasons. It is a top-ranked and downloaded resource at Hugging Face, a provider of NLP resources⁷. A diversity of sentences are continually being added to the database, which can be used to train and update the training of VAs and other NLP systems. As an open data set, it has a CC BY 2.0 FR licence, though some sentences are public domain. Notably, a portion of the English and Japanese sentences were sourced from the public domain Tanaka Corpus [40]. As a big data set, it has 255,675 sentences at the time of writing. It is an established and active crowdsourcing project, with participants around the world.

3.2 Data Preparation and Analysis

All data analyses were performed in Python (version 3.8.10) with the Tatoebatools (version 0.2.1) package⁸. This package allowed us to download the Tatoeba data sets in Python. The English data set was tokenized, i.e., sentences were divided into separate words/terms, and tagged according to their linguistic meaning, e.g., nouns, pronouns, verbs, etc., using the Natural Language Toolkit (nltk) package (version 3.8.1). The Japanese data set was tokenized and tagged

⁴<https://osf.io/8jmyc>

⁵https://tatoeba.org/en/stats/native_speakers

⁶https://tatoeba.org/en/stats/users_languages

⁷<https://huggingface.co/datasets/tatoeba>

⁸<https://pypi.org/project/tatoebatools>

³<https://arxiv.org/abs/2204.08582>

using the Janome⁹ (version 0.4.2) package. For data preprocessing, we filtered out Japanese special characters, such as full-width spaces, ‘, ‘, ‘, etc. Then we extracted pronouns from the English sentences using the pronoun tags created by nltk, namely the PRP and PRP\$ tags, and the 代名(dai-mei-shi) tag created by Janome from the Japanese sentences (refer to 3.3 for details). We identified the genderedness of the Japanese pronouns by referring to Jisho.org¹⁰, which is an online Japanese dictionary that uses Tatoeba to provide example sentences between Japanese and English. We developed three pronoun categories: *masculine*, *feminine*, and, to account for pronoun ambiguity [41], *ambiguous*, including singular first-person pronouns, singular and/or plural “they/them” pronouns, words that may or may not be pronouns, such as 何(nan; can refer to a person or “what”), and gender-diverse pronouns used by genderqueer people, such as “xim/xir,” that could be pronouns or represent truncated words or typos [32], which only manual analysis could confirm.

We generated descriptive statistics, i.e., means (M) and standard deviation (SD) of counts and ratios/percentages. We created a confusion matrix (or error matrix), which maps the relative distribution of one or more variables before and after algorithmic treatment [37] and is often used for automated content analysis and classification in machine learning (ML). We used inferential statistics, i.e., Chi-square tests, to compare groups, i.e., pronoun genders, languages, matching or not matching translations.

3.3 Pronoun Tags

We used the pronoun dictionary provided by the nltk package’s Part of Speech (POS)¹¹ library for the English data set, and the tokenizer created by Kudo, Yamamoto, and Matsumoto [24] for the Japanese data set. However, because these libraries did not represent an up-to-date or inclusive spread of pronouns, we manually added several pronouns according to recent NLP work [25]. The lists of pronouns that we used are as follows:

Japanese masculine:

きみ、君、お前、俺、彼、彼ら、君たち、オレ、おまえ、ぼく、ボク、ら、おれ、吾、キミ、てめえ、小生、てめえ、たち

kimi, kun, omae, ore, kare, karera, boku, kuntachi, ore, omae, boku, boku, bokura, bokutachi, ore, wagahai, kimi, temē, shōsei, temē, bokutachi

Japanese feminine:

彼女、あたし、彼女ら

kanojo, atashi, kanojora

Japanese ambiguous:

何、私、それ、あなた、みんな、あいつ、わたし、方、どなた、我々、あんた、そつ、やつ、みな、奴、あれ、なに、皆、みなさん、みなさま、我ら、余、彼等、だれ、奴ら、うち、わたくし、われわれ、よそ、われ、奴等、己、おのれ、何れ、わし、彼奴

nani, watashi, sore, anata, minna, aitsu, dare, watashi, anata, donata, wareware, anta, so, yatsu, mina, yakko,

⁹<https://mocabeta.github.io/janome>

¹⁰<https://jisho.org>

¹¹<https://www.nltk.org/book/ch05.html>

Table 1: Descriptive statistics by pronoun category and language.

Gender Category	English	Japanese
Masculine	43,453	50,350
Feminine	19,025	17,123
Ambiguous	148,903	61,916
Non-Masculine	167,928	79,039

are, nani, mina, minasan, minasama, warera, yo, karera, dare, yatsura, uchi, watakushi, wareware, yoso, ware, yatsura, onore, onore, izure, washi, aitsu

English masculine: he, him, his, himself

English feminine: her, she, herself

English ambiguous:

I, you, it, me, my, your, them, their, myself, they, themselves, we, us, oneself, our, yourself, its, itself, self, ourselves, 'em, theirs, thyself, one, ours, themselves, theirself, theirselves, theirselves, xe, xem, xim, hir, xemself, xemselves, hirsself, hirselves, ze, zeself, zeselves

4 RESULTS

We now turn to answering the hypotheses following data analysis.

4.1 H1. There will be mismatches in direct pronoun translations from English to Japanese.

A Chi-square test was run to see if the number of pronouns by pronoun category (Table 1) differed between the English and Japanese data sets, i.e., because of the translation process, where pronouns were classified into three categories: masculine (EN = 43,453, JP = 50,350), feminine (EN = 19,025, JP = 17,123), and ambiguous (EN = 148,903, JP = 61,916). A medium, statistically significant relationship was found, $\chi^2(2, N = 340,770) = 17,802.0, p < .001$, Cramer’s $V = .23$. Overall, this confirms that a significant change occurred through the translation process, so we can accept H1.

4.2 H2: There will be a masculine bias in pronoun use in the data sets.

Chi-square tests with Yates corrections were run to compare masculine and non-masculine pronouns by language. A statistically significant difference comparing masculine and feminine pronouns to ambiguous pronouns was found for the English data set, $\chi^2(1, N=360,284) = 7930.40, p < .001$, Cramer’s $V = .15$, and the Japanese data set, $\chi^2(1, N=208,428) = 17,300.46, p < .001$, Cramer’s $V = .29$. Descriptive statistics (Table 1) indicate that there were more masculine pronouns used regardless of language compared to feminine pronouns. Another Chi-square test was run to see if the number of masculine (EN = 43,453, JP = 50,350) and non-masculine (EN = 167,928, JP = 79,039) pronouns differed between the English and Japanese data sets. A weak, statistically significant relationship was found, $\chi^2(1, N= 340,770) = 13,557.2, p < .001$, Cramer’s $V = .20$. Descriptive statistics (Table 1) indicate that use of masculine

Table 2: Descriptive statistics for matching and mismatching translations.

Gender Category	Match	Mismatch
Masculine	36,164	21,475
Non-masculine	71,527	103,913
Feminine	16,855	2,438
Non-feminine	90,836	122,950

pronouns increased in the Japanese data set, while use of other pronouns decreased. Overall, a significant difference in representation between masculine and non-masculine pronouns was found, apparently because of the translation process, supporting H2.

4.3 H3: There will be a masculine bias in pronoun translation from English to Japanese.

We considered whether the number of masculine (Match = 36,164, Mismatch = 21,475) and non-masculine (Match = 71,527, Mismatch = 103,913) pronouns differed based on translation mis/matching between English and Japanese. A Chi-square test revealed a low but statistically significant relationship between these two variables, $\chi^2(1, N = 233,079) = 8,426.7, p < .001$, Cramer's $V = .19$. As a comparison, we then considered whether the number of feminine (Match = 16,855, Mismatch = 2,438) and non-feminine (Match = 90,836, Mismatch = 122,950) pronouns based on mis/matching between Japanese and English. mA Chi-square test revealed a low but statistically significant relationship, $\chi^2(1, N = 233,079) = 14,336.3, p < .001$, Cramer's $V = .25$. The degree of feminine pronouns compared to other pronouns, in light of the results for masculine pronouns, indicates a higher interchange between ambiguous and masculine pronouns, i.e., a masculine bias, supporting H3. Descriptive statistics (Table 2) indicate that use masculine pronouns decreased, i.e., were changed to other pronouns or removed on translation, thus leading us to reject H3.

We created a confusion matrix (Figure 1) to clarify the interchange of English and Japanese pronouns, i.e., the extent to which

each pronoun or combination of pronouns per sentence matched on translation. The diagonal values (in green) represent the cases where English and Japanese matched (only 57% of all pronouns). The top-right quadrant (in orange) represents the Japanese data, and the bottom-left quadrant (in purple) represents the English data. The largest difference was for ambiguous pronouns (English ambiguous, Japanese none = 73,228). The next largest difference was between English ambiguous pronouns and (i) Japanese masculine pronouns ($n = 11,200$) and (ii) Japanese sentences that include both masculine and ambiguous pronouns ($n = 2,324$). Both indicate a masculine bias in the Japanese data set. Comparatively, conversions between ambiguous and (i) feminine and (ii) sentences with feminine and ambiguous pronouns was only $n = 96$ and $n = 50$, respectively. Feminine pronouns scored the highest, which means that feminine pronouns were specified and excluded, while neutral and masculine pronouns were removed or used interchangeably. Finally, the matrix visualizes the masculine bias in matched cases between Japanese and English ($n = 21,798$).

5 DISCUSSION

Our findings reveal clear gender biases in pronouns for Tatoeba, notably masculine biases in the English and Japanese data sets. Moreover, there were mistranslations, and not only with respect to ambiguous pronouns. Furthermore, these mistranslations appear to have been driven by the relative presence and absence of other pronouns. We discuss these nuances and propose a fix for embracing pronoun plurality in Tatoeba and other NLP data sets.

5.1 Nuances in Mistranslations: Sources of Masculine Biases and Pronoun Missteps

A significant number of mismatching pronoun translations alongside an overrepresentation of masculine pronouns was found in Tatoeba, especially in the Japanese data set. Tatoeba translation can go either way, so we cannot be sure in which language these biases originated; we can only acknowledge the ground truth of the crowdsourced translation method itself. Even so, the results point to an increase in masculine pronouns in the Japanese data set, highlighted against the relatively low levels of feminine pronouns

Matched		Japanese								
Categories	None	A	F	F,A	M	M,A	F,M	F,M,A		
English	None	60204	5171	39	5	129	21	1	0	Highest
	A	73328	42657	96	50	11220	2324	25	3	
	F	1242	48	9771	409	2	0	12	0	
	F,A	421	221	1903	2471	51	7	372	19	
	M	4504	476	15	1	21798	1041	16	0	Lowest
	M,A	1230	827	5	1	5440	6012	8	3	
	F,M	76	5	101	10	57	6	1561	51	
	F,M,A	19	7	7	5	6	2	100	63	
M: Masculine		F: Feminine		A: Ambiguous						

Figure 1: Confusion matrix showing the distribution of mis/matched pronoun translations. Counts by English pronoun categories (purple) are contrasted with Japanese pronoun categories (orange); matched cases fall along the diagonal line (green). Lighter tones indicate lower counts, while darker tones indicate higher counts. None: Non-gendered pronouns, i.e., “it,” names, and non-humans.

and mis/matched translations. One possibility is the presence of masculine language biases in Japanese [26]; yet, this is also true for English [2]. Without research directly comparing Japanese and English outside of translation and this small context of Tatoeba, it is hard to draw conclusions about whether we should expect either language to be more masculine-biased than the other. However, the case of ambiguous pronouns provides some nuance. Most were simply absent in the Japanese data set, likely due to the number of neutral pronouns such as “I” and “you” in English, which are omitted in Japanese, and “they/them,” which can be a non-binary singular pronoun or a gender-neutral plural pronoun. Even so, the high number of masculine pronouns alongside ambiguous pronouns in the Japanese data set, where ambiguous pronouns are used in English, points to translation bias. Nevertheless, we can feel some confidence in the sheer number of ambiguous pronouns, which leave open the “genderedness” of language to the reader/listener and especially allow for queer readings of the text.

5.2 A Pluralistic Approach to Pronoun Representation in Translated Data Sets

One of the most puzzling aspects of this work, particular to databases structured like Tatoeba, is the support of plurality for other language matters. Specifically, Tatoeba will repeat sentences in one or the other language to allow for a plurality of linguistic expressions, e.g., terms and grammar, and multiple translations of that sentence. For example:

Line 49: 生物学は好きになれません。

Line 50: 私は生物学はして好きではありませんでした。

Line 49: I never liked biology.

Line 50: I never liked biology.

We might think to do the same for pronouns. However, there are a lot of pronouns, so this seems infeasible; the more lines added to the database, the more baseline repetition and the greater the file size of the database. We suggest another approach, drawing from similar efforts in NLP data sets: standardized labels or tags. In MASSIVE, for instance, there are tags for “time” and “date,” as in the example below:

```
{ "id": "0", "locale": "en-US", "partition": "test", "scenario": "alarm", "intent": "alarm_set", "utt": "wake me up at five am this week", "annot_utt": "wake me up at [time : five am] [date : this week]", "worker_id": "1" }
```

Here, the tags are merely annotating the data, rather than acting as variables that could replace the data. We suggest going one step further. A pronoun tag or placeholder would take the place of the text. More like a variable, it could refer to a list of pronouns appropriate for the sentence. To rework an example from Tatoeba:

Line 6936: 味があるなんてものではなく、もう中なんです。

Line 6936: [p] is not just interested, [p]’s crazy about it.

There may be multiple lists depending on the context. For instance, there may be cases where a famous person or character with known

pronouns is being referenced, and so to avoid misgendering that person, a special placeholder referring to certain subset of the pronoun list would be used. Future work can map out a formal specification.

We recognize that this would mean moving away from a plain text approach to a structured format, which we admit has implications for a project as large as Tatoeba; yet, we argue that, as a matter of inclusion and accuracy, this should be done. Still, we should consider how contributors would make contributions. Practically, it would be a simple matter of requiring a placeholder for pronouns rather than a particular pronoun within each sentence. Contributors could use a generic placeholder, such as [p] (for pronoun), which admins, corpus maintainers, and advanced contributors could then assess and revise, as needed. Still, we cannot expect to inform or train every contributor in a crowdsourcing initiative about new structures and rules. A future project could be the development of an AI-driven interface for inputting sentences into Tatoeba, one that is trained on the pronoun lists and watches for these pronouns, and either suggests to the contributor that they be replaced with a placeholder like [p] or automatically replaces them.

5.3 Limitations

We used one database (Tatoeba) and two languages (English and Japanese). While we provided reasons for this, we urge future work to evaluate whether and how the findings “translate” to other databases and other languages. Moreover, Tatoeba differs from other data sets in that the translation can go either way, so we could not determine the source of mistranslations. Future work will need to evaluate data sets with a known source language.

6 CONCLUSION

Human language is biased, and so is the human mind. We have revealed how such biases crop up in translations related to gender through the case study of pronouns in English and Japanese sentences from Tatoeba, an established, large-scale, open, crowdsourced translation project for NLP and used for training VAs. We aim to raise awareness, but we have also provided a fix that could be adopted by Tatoeba and other initiatives. Future work will need to explore the presence and extent of these “translation missteps” in other data sets and the feasibility of our proposed solution.

ACKNOWLEDGMENTS

This work was funded in part by a Tokyo Tech Young Investigator Engineering Award.

REFERENCES

- [1] April H. Bailey, Marianne LaFrance, and John F. Dovidio. 2019. Is man the measure of all things? A social cognitive account of androcentrism. *Personality and Social Psychology Review* 23, 4, 307-331. <https://doi.org/10.1177/1088868318782848>
- [2] April H. Bailey, Adina Williams, and Andrei Cimpian. 2022. Based on billions of words on the internet, people = men. *Science Advances* 8, 13, eabm2463. <https://doi.org/10.1126/sciadv.abm2463>
- [3] Shaowen Bardzell. 2010. Feminist HCI: Taking stock and outlining an agenda for design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '10), 1301-1310. <https://doi.org/10.1145/1753326.1753521>
- [4] Shaowen Bardzell and Jeffrey Bardzell. 2011. Towards a feminist HCI methodology: Social science, feminism, and HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '11), 675-684. <https://doi.org/10.1145/1978942.1979041>
- [5] Simone de Beauvoir. 2011. *The Second Sex*. Vintage Books, New York, NY.

- [6] Rosanna Bellini, Angelika Strohmayer, Ebtisam Alabulqader, Alex A. Ahmed, Katta Spiel, Shaowen Bardzell, and Madeline Balaam. 2018. Feminist HCI: Taking stock, moving forward, and engaging community. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (CHI EA '18), 1–4. <https://doi.org/10.1145/3170427.3185370>
- [7] Emily M. Bender and Batya Friedman. 2018. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics* 6, 587–604. https://doi.org/10.1162/tacl_a_00041
- [8] Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. 2020. Language (technology) is power: A critical survey of “bias” in NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (ACL 2020), 5454–5476. <https://doi.org/10.18653/v1/2020.acl-main.485>
- [9] Su Lin Blodgett, Q. Vera Liao, Alexandra Olteanu, Rada Mihalcea, Michael Muller, Morgan Klaus Scheuerman, Chenhao Tan, and Qian Yang. 2022. Responsible language technologies: Foreseeing and mitigating harms. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI EA '22), 1–3. <https://doi.org/10.1145/3491101.3516502>
- [10] Yang Trista Cao and Hal Daumé III. 2021. Toward gender-inclusive coreference resolution: An analysis of gender and bias throughout the machine learning lifecycle. *Computational Linguistics* 47, 3, 615–661. https://doi.org/10.1162/coli_a_00413
- [11] Tommaso Caselli, Roberto Cibir, Costanza Conforti, Enrique Encinas, and Maurizio Teli. 2021. Guiding principles for participatory design-inspired natural language processing. In *Proceedings of the 1st Workshop on NLP for Positive Impact* (ACL-IJCNLP-NLP4PosImpact 2021), 27–35. <https://doi.org/10.18653/v1/2021.nlp4posimpact-1.4>
- [12] Yan Chen, Christopher Mahoney, Isabella Grasso, Esma Wali, Abigail Matthews, Thomas Middleton, Mariama Njie, and Jeanna Matthews. 2021. Gender bias and under-representation in natural language processing across human languages. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society* (AI/ES '21), 24–34. <https://doi.org/10.1145/3461702.3462530>
- [13] Shruthi Sai Chivukula and Colin M. Gray. 2020. Bardzell’s “Feminist HCI” legacy: Analyzing citational patterns. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI EA '20), 1–8. <https://doi.org/10.1145/3334480.3382936>
- [14] Won Ik Cho, Ji Won Kim, Seok Min Kim, and Nam Soo Kim. 2019. On measuring gender bias in translation of gender-neutral pronouns. In *Proceedings of the First Workshop on Gender Bias in Natural Language Processing* (GeBNLP 2019), 173–181. <https://doi.org/10.18653/v1/W19-3824>
- [15] Leigh Clark, Philip Doyle, Diego Garaialde, Emer Gilmartin, Stephan Schlögl, Jens Edlund, Matthew Aylett, João Cabral, Cosmin Munteanu, Justin Edwards, and Benjamin R. Cowan. 2019. The state of speech in HCI: Trends, themes and challenges. *Interacting with Computers* 31, 4, 349–371. <https://doi.org/10.1093/iwc/iwz016>
- [16] Mar Díaz-Millón and María Dolores Olvera-Lobo. 2021. Towards a definition of transcreation: a systematic literature review. *Perspectives* 0, 0, 1–18. <https://doi.org/10.1080/0907676X.2021.2004177>
- [17] Emily Dinan, Angela Fan, Ledell Wu, Jason Weston, Douwe Kiela, and Adina Williams. 2020. Multi-dimensional gender bias classification. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP) (EMNLP 2020), 314–331. <https://doi.org/10.18653/v1/2020.emnlp-main.23>
- [18] Anjalie Field and Yulia Tsvetkov. 2020. Unsupervised discovery of implicit gender bias. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP) (EMNLP 2020), 596–608. <https://doi.org/10.18653/v1/2020.emnlp-main.44>
- [19] Jack FitzGerald, Christopher Hench, Charith Peris, Scott Mackie, Kay Rottmann, Ana Sanchez, Aaron Nash, Liam Urbach, Vishesh Kakarala, and Richa Singh. 2022. MASSIVE: A 1M-example multilingual natural language understanding dataset with 51 typologically-diverse languages. *arXiv preprint arXiv:2204.08582*.
- [20] Ismael Garrido-Muñoz, Arturo Montejo-Ráez, Fernando Martínez-Santiago, and L. Alfonso Ureña-López. 2021. A survey on bias in deep NLP. *Applied Sciences* 11, 7, 3184. <https://doi.org/10.3390/app11073184>
- [21] Liane Guillou and Christian Hardmeier. 2016. PROTEST: A test suite for evaluating pronouns in machine translation. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation* (LREC'16) (LREC 2016), 636–643. Retrieved January 20, 2023 from <https://aclanthology.org/L16-1100>
- [22] Samantha Jaroszewski, Danielle Lottridge, Oliver L. Haimson, and Katie Quehl. 2018. “Genderfluid” or “attack helicopter”: Responsible HCI research practice with non-binary gender variation in online communities. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (CHI '18), 1–15. <https://doi.org/10.1145/3173574.3173881>
- [23] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (ACL 2020), 6282–6293. <https://doi.org/10.18653/v1/2020.acl-main.560>
- [24] Taku Kudo, Kaoru Yamamoto, and Yuji Matsumoto. 2004. Applying conditional random fields to Japanese morphological analysis. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing* (EMNLP 2004), 230–237. Retrieved January 20, 2023 from <https://aclanthology.org/W04-3230>
- [25] Anne Lauscher, Archie Crowley, and Dirk Hovy. 2022. Welcome to the modern world of pronouns: Identity-inclusive natural language processing beyond gender. In *Proceedings of the 29th International Conference on Computational Linguistics* (COLING 2022), 1221–1232. Retrieved November 23, 2022 from <https://aclanthology.org/2022.coling-1.105>
- [26] Jackie F.K. Lee. 2018. Gender representation in Japanese EFL textbooks – a corpus study. *Gender and Education* 30, 3, 379–395. <https://doi.org/10.1080/09540253.2016.1214690>
- [27] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebu. 2019. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–229. <https://doi.org/10.1145/3287560.3287596>
- [28] Christine Murad, Cosmin Munteanu, Benjamin R. Cowan, Leigh Clark, Martin Porcheron, Heloisa Candelero, Stephan Schlögl, Matthew P. Aylett, Jaisie Sin, Robert J. Moore, Grace Hughes, and Andrew Ku. 2021. Let’s talk about CUIs: Putting conversational user interface design into practice. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (CHI EA '21), 1–6. <https://doi.org/10.1145/3411763.3441336>
- [29] Tsuyoshi Ono and Sandra A. Thompson. 2003. Japanese (w)atashi/ore/boku I: They’re not just pronouns. *Cognitive Linguistics* 14, 4, 321–347. <https://doi.org/10.1515/cogl.2003.013>
- [30] Marcelo O. R. Prates, Pedro H. Avelar, and Luis C. Lamb. 2020. Assessing gender bias in machine translation: A case study with Google Translate. *Neural Computing and Applications* 32, 10, 6363–6381. <https://doi.org/10.1007/s00521-019-04144-6>
- [31] Ari Schlesinger, W. Keith Edwards, and Rebecca E. Grinter. 2017. Intersectional HCI: Engaging identity through gender, race, and class. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (CHI '17), 5412–5427. <https://doi.org/10.1145/3025453.3025766>
- [32] Katie Seaborn, Shruti Chandra, and Thibault Fabre. 2023. Transcending the “male code”: Implicit masculine biases in NLP contexts. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23), 1–19. <https://doi.org/10.1145/3544548.3581017>
- [33] Katie Seaborn and Alexa Frank. 2022. What pronouns for Pepper? A critical review of gender/ing in research. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI '22), 1–15. <https://doi.org/10.1145/3491102.3501996>
- [34] Katie Seaborn, Norihisa Paul Miyake, Peter Pennefather, and Mihoko Otake-Matsuura. 2022. Voice in human-agent interaction: A survey. *ACM Computing Surveys* (CSUR) 54, 4, Article No. 81. <https://doi.org/10.1145/3386867>
- [35] Katta Spiel, Oliver L. Haimson, and Danielle Lottridge. 2019. How to do better with gender on surveys: A guide for HCI researchers. *Interactions* 26, 4, 62–65. <https://doi.org/10.1145/3338283>
- [36] Damiano Spina, Johanne R. Trippas, Paul Thomas, Hideo Joho, Katriina Byström, Leigh Clark, Nick Craswell, Mary Czerwinski, David Elswiler, Alexander Frummet, Souvick Ghosh, Johannes Kiesel, Irene Lopatovska, Daniel McDuff, Selina Meyer, Ahmed Mourad, Paul Owiocho, Sachin Pathiyan Cherumanal, Daniel Russell, and Laurianne Sitbon. 2021. Report on the future conversations workshop at CHIIR 2021. *ACM SIGIR Forum* 55, 1, 6:1–6:22. <https://doi.org/10.1145/3476415.3476421>
- [37] Stephen V. Stehman. 1997. Selecting and interpreting measures of thematic classification accuracy. *Remote Sensing of Environment* 62, 1, 77–89. [https://doi.org/10.1016/S0034-4257\(97\)00083-7](https://doi.org/10.1016/S0034-4257(97)00083-7)
- [38] Yolande Strengers, Lizhen Qu, Qiongkai Xu, and Jarrod Knibbe. 2020. Adhering, steering, and queering: Treatment of gender in natural language generation. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI '20), 1–14. <https://doi.org/10.1145/3313831.3376315>
- [39] Tony Sun, Andrew Gaut, Shirlyn Tang, Yuxin Huang, Mai ElSherief, Jieyu Zhao, Diba Mirza, Elizabeth Belding, Kai-Wei Chang, and William Yang Wang. 2019. Mitigating gender bias in natural language processing: Literature review. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (ACL 2019), 1630–1640. <https://doi.org/10.18653/v1/P19-1159>
- [40] Yasuhito Tanaka. 2001. Compilation of a multilingual parallel corpus. In *Proceedings of the 2001 International Conference of the Pacific Association for Computational Linguistics* (PACLING 2001) (PACLING 2001), 265–268.
- [41] Kellie Webster, Marta Recasens, Vera Axelrod, and Jason Baldridge. 2018. Mind the GAP: A balanced corpus of gendered ambiguous pronouns. *Transactions of the Association for Computational Linguistics* 6, 605–617. https://doi.org/10.1162/tacl_a_00240
- [42] James Zou and Londa Schiebinger. 2018. *AI can be sexist and racist—it’s time to make it fair*. Nature Publishing Group. Retrieved from <https://www.nature.com/articles/d41586-018-05707-8>